C

LINEAR ALGEBRA • UNIT 4.01

Determinants & Geometric Volume

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 C	Q9 B	Q10 D
Q11 A	Q12 C	Q13 B	Q14 A	Q15 C
Q16 D	Q17 D	Q18 B	Q19 B	Q20 D
Q21 A	Q22 B	Q23 C	Q24 B	Q25 B

Determinants & Geometric Volume Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C 15 ✓

D 225

WORKED SOLUTION & VERIFICATION

$\|\mathbf{u}\| = \sqrt{(-9)^2 + (-12)^2} = \sqrt{225} = 15.$

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C 29 ✓

D 41

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 29$.

3 MCQ

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

- **A** $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- **C** $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
- **D** $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

- **A** (7, 6) ✓
- **B** (6, 7)
- **C** (8, 5)
- **D** (5, 8)

WORKED SOLUTION & VERIFICATION

The transformation is defined by the matrix multiplication $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$. We are given the point $(3, 1)$, so $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Performing the matrix multiplication:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (1 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 6 + 1 \\ 3 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

- **A** 1
- **B** 0 ✓
- **C** -7
- **D** 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

We are given the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2-2(1) & -1-2(2) & 0-2(5) \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right)$$

Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right)$$

This matrix is now in row-echelon form. The corresponding system of equations is:

$x + 2y = 5$

$y = 2$

Substitute the value of y into the first equation:

$x + 2(2) = 5$

$x + 4 = 5$

$x = 1$

Therefore, the unique solution is $(1, 2)$, and the value of x is 1.

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

- **A** 13 ✓
- **B** 17
- **C** 7
- **D** 169

WORKED SOLUTION & VERIFICATION

The magnitude of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$, we have $x = 5$ and $y = -12$. Therefore, $\|\mathbf{v}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C **The distance of the point (3,4) from the origin.** ✓

D The sum of the components of the vector.

WORKED SOLUTION & VERIFICATION

The length of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by its magnitude, $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 MCQ

Consider a parallelepiped in R^3 formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.

A 3

B 2 ✓

C 4

D 6

WORKED SOLUTION & VERIFICATION

The volume of a parallelepiped formed by three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ is given by the absolute value of their scalar triple product, which can be computed as the absolute value of the determinant of the matrix formed by these vectors as columns (or rows).

Let the matrix be $M = (2 \quad 1 \quad 0\ 1 \quad 3 \quad 2\ k \quad 0 \quad 4)$.

The determinant of M is calculated as:

$\det(M) = 2\begin{vmatrix} 3 & 2\ 0 \\ 4 \end{vmatrix} - 1\begin{vmatrix} 1 & 2\ k \\ 4 \end{vmatrix} + 0\begin{vmatrix} 1 & 3\ k \\ 0 \end{vmatrix}$

$\det(M) = 2((3)(4) - (2)(0)) - 1((1)(4) - (2)(k)) + 0$

$\det(M) = 2(12) - (4 - 2k)$

$\det(M) = 24 - 4 + 2k$

$\det(M) = 20 + 2k$

The volume of the parallelepiped is given as 24 cubic units. Therefore, $|\det(M)| = 24$.

$|20 + 2k| = 24$.

This gives two possibilities:

1. $20 + 2k = 24$

$2k = 4$

$k = 2$

2. $20 + 2k = -24$

$2k = -44$

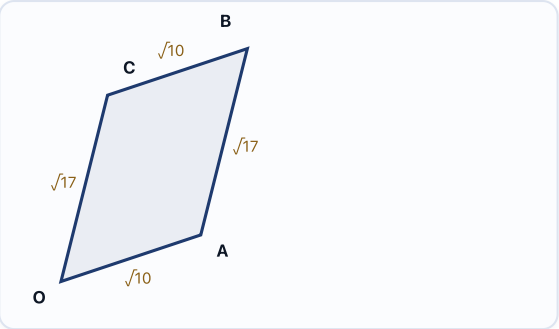
$k = -22$

Since the problem states that $k > 0$, the correct value for k is 2.

Common Algebraic Trap: Students might forget to take the absolute value of the determinant, or they might make errors in the cofactor expansion. If $k = -22$ was chosen, it would be incorrect due to the condition $k > 0$.

10 MCQ

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. What is the area of this parallelogram?



A 10

B 13

C 14

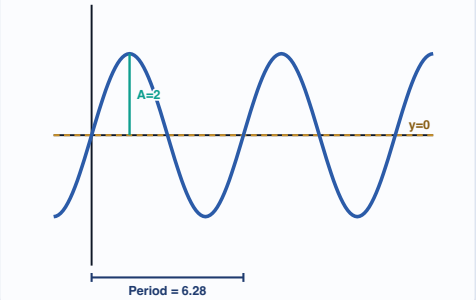
D **11** ✓

WORKED SOLUTION & VERIFICATION

The area of a parallelogram defined by two vectors $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ in R^2 is given by the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). In this case, the matrix is $A = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}$. The determinant of A is $\det(A) = (3)(4) - (1)(1) = 12 - 1 = 11$. Therefore, the area of the parallelogram is $|11| = 11$.

11 MCQ

A solid object in R^3 is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathrm{R}^3 \to \mathrm{R}^3$ is represented by the matrix $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, what is the volume of the transformed solid object?



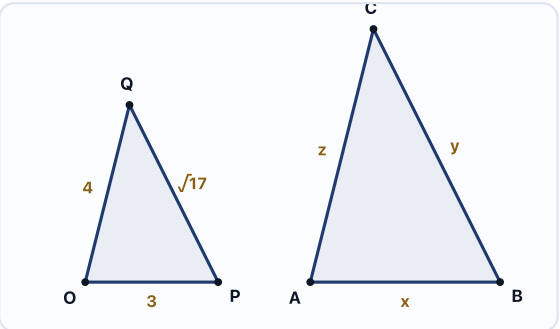
- **A** 24 ✓
- **B** 12
- **C** 36
- **D** 48

WORKED SOLUTION & VERIFICATION

The volume of the solid region bounded by the coordinate planes and the plane passing through $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$ can be calculated. The equation of the plane passing through these points is given by $\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$. This is a tetrahedron with vertices at $(0, 0, 0)$, $(6, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 3)$. The volume of such a tetrahedron is $V_{original} = \frac{1}{6}|abc|$, where a, b, c are the intercepts on the axes. Thus, $V_{original} = \frac{1}{6}|6 \cdot 4 \cdot 3| = \frac{1}{6}|72| = 12$ cubic units. The linear transformation T is represented by the matrix M. The volume of the transformed solid object is given by $V_{transformed} = |\det(M)| \cdot V_{original}$. The determinant of M is $\det(M) = 1 \cdot (1 \cdot 2 - 0 \cdot 0) - 0 + 0 = 2$. Therefore, $V_{transformed} = |2| \cdot 12 = 24$ cubic units.

12 MCQ

A parallelogram in the xy-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?



- **A** 13
- **B** 15
- **C** 11 ✓
- **D** 17

WORKED SOLUTION & VERIFICATION

The area of a parallelogram formed by vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ originating from the same point is given by the absolute value of the determinant of the matrix formed by these vectors: $|\det([u_1 \quad v_1 \; u_2 \quad v_2])|$. \n\nIn this case, we can take the vectors originating from $(0, 0)$ to be $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 4)$. \n\nThe area is $|\det([3 \quad 1 \; 1 \quad 4])| = |(3 \times 4) - (1 \times 1)| = |12 - 1| = |11| = 11$.

13 MCQ

Consider the parallelepiped in R^3 defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$ with $k > 3$, and $\mathbf{w} = (3, 4, 1)$. If the volume of this parallelepiped is $\frac{45}{2}$ cubic units, find the value of k.

Given Vector Components

Component x	Component y	Component z
2	1	-1

A $\frac{19}{2}$

B $\frac{15}{2}$ ✓

C $\frac{13}{2}$

D $\frac{17}{2}$

WORKED SOLUTION & VERIFICATION

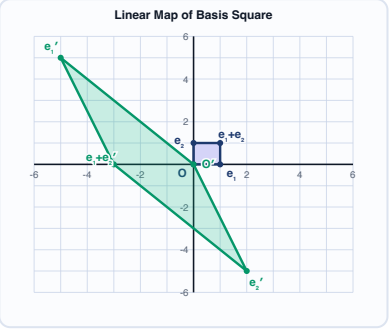
The volume of the parallelepiped is the absolute value of the determinant of the matrix formed by $\mathbf{u}, \mathbf{v}, \mathbf{w}$:

$$\det \begin{pmatrix} 2 & 1 & -1 \\ 1 & k & 2 \\ 3 & 4 & 1 \end{pmatrix} = 2(k-8) - 1(1-6) - 1(4-3k) = 5k - 15.$$

For $k > 3$, $5k - 15 = \frac{45}{2} \implies 5k = \frac{75}{2} \implies k = \frac{15}{2}$.

14 MCQ

For $A = \begin{bmatrix} -5 & 2 \\ 5 & -5 \end{bmatrix}$, find entry $(2A)_{22}$.



A -10 ✓

B -5

C 10

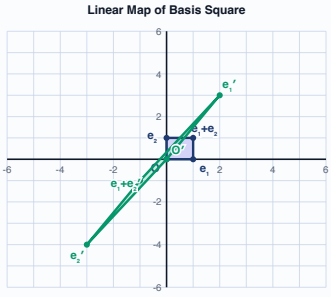
D -9

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(2A)_{22} = 2(-5) = -10$. Therefore, the verified correct option is -10.

15 MCQ

For $A = \begin{bmatrix} 2 & -3 \\ 3 & -4 \end{bmatrix}$, find entry $(-4A)_{21}$.



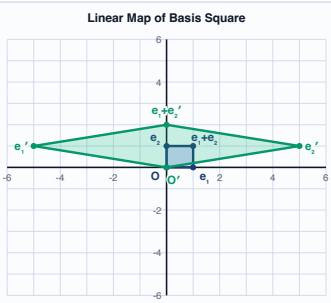
A 3	B 12
C -12 ✓	D -11

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-4A)_{21} = -4(3) = -12$. Therefore, the verified correct option is -12.

16 MCQ

For $A = \begin{bmatrix} -5 & 5 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -6 \\ -2 & -4 \end{bmatrix}$, find entry $(A+B)_{11}$.



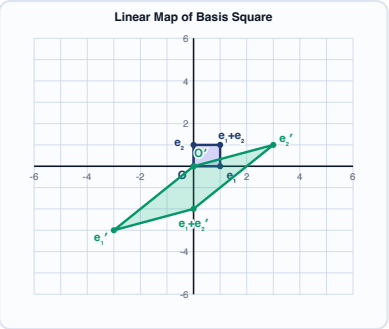
A -8	B -5
C -1	D -6 ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{11} = (-5) + (-1) = -6$. Therefore, the verified correct option is -6.

17 MCQ

For $A = \begin{bmatrix} -3 & 3 \\ -3 & 1 \end{bmatrix}$, find entry $(4A)_{22}$.



A 5

B 1

C -4

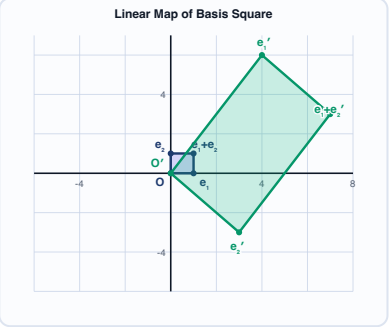
D 4 ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(4A)_{22} = 4(1) = 4$. Therefore, the verified correct option is 4.

18 MCQ

For $A = \begin{bmatrix} 4 & 3 \\ 6 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -6 \\ 5 & 4 \end{bmatrix}$, find entry $(A+B)_{11}$.



A 4

B 9 ✓

C 5

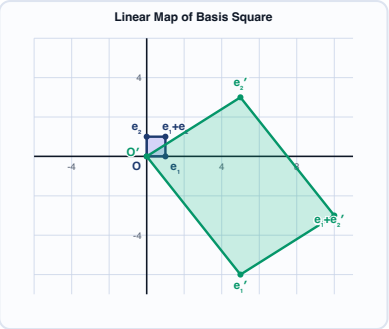
D 10

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{11} = (4) + (5) = 9$. Therefore, the verified correct option is 9.

19 MCQ

For $A = \begin{bmatrix} 5 & 5 \\ -6 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 5 \\ -5 & -5 \end{bmatrix}$, find entry $(A+B)_{21}$.



A -10

B **-11** ✓

C -6

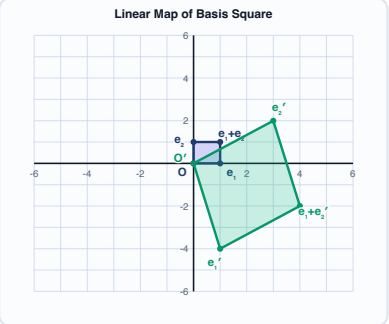
D -5

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{21} = (-6)+(-5) = -11$. Therefore, the verified correct option is -11.

20 MCQ

For $A = \begin{bmatrix} 1 & 3 \\ -4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ -2 & -2 \end{bmatrix}$, find row 1 of AB.



A [7, -7]

B [-8, -7]

C [-1, -6]

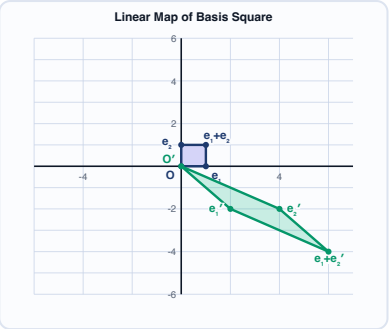
D **[-7, -8]** ✓

WORKED SOLUTION & VERIFICATION

$(AB)_{11} = 1(-1)+(3)(-2) = -7$, $(AB)_{12} = 1(-2)+(3)(-2) = -8$.

21 MCQ

For $A = \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, find row 2 of AB.



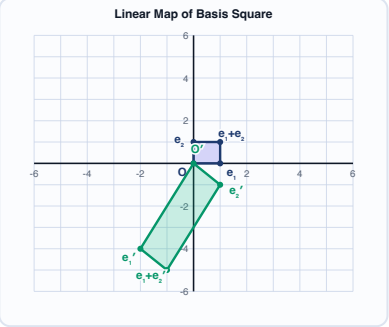
- **A** [0, 0] ✓
- **B** [-4, -6]
- **C** [-2, -2]
- **D** [-2, 0]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = -2(1) + (-2)(-1) = 0$, $(AB)_{22} = -2(-1) + (-2)(1) = 0$.

22 MCQ

For $A = \begin{bmatrix} -2 & 1 \\ -4 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ -4 & 2 \end{bmatrix}$, find row 2 of AB.



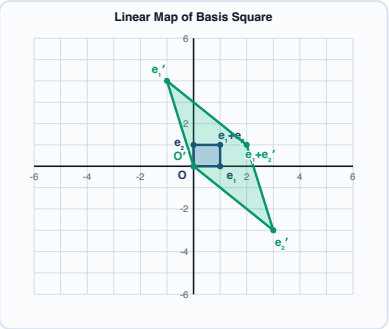
- **A** [0, -6]
- **B** [0, -18] ✓
- **C** [-18, 0]
- **D** [-4, -2]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = -4(1) + (-1)(-4) = 0$, $(AB)_{22} = -4(4) + (-1)(2) = -18$.

23 MCQ

For $A = \begin{bmatrix} -1 & 3 \\ 4 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & -2 \\ 2 & -2 \end{bmatrix}$, find row 2 of AB.



A [-10, 12]

B [-2, -10]

C **[-10, -2]** ✓

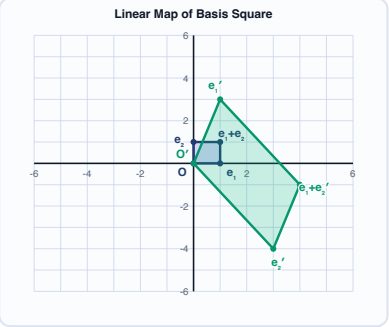
D [-4, 6]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = 4(-1) + (-3)(2) = -10$, $(AB)_{22} = 4(-2) + (-3)(-2) = -2$.

24 MCQ

For $A = \begin{bmatrix} 1 & 3 \\ 3 & -4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix}$, find row 2 of AB.



A [2, -7]

B **[-2, -7]** ✓

C [-7, -2]

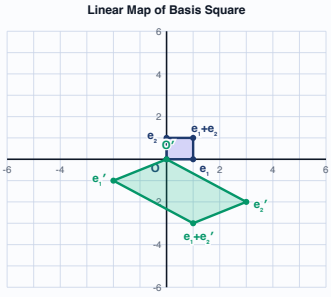
D [-6, -4]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = 3(-2) + (-4)(-1) = -2$, $(AB)_{22} = 3(-1) + (-4)(1) = -7$.

25 MCQ

For $A = \begin{bmatrix} -2 & 3 \\ -1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -1 \\ -2 & 3 \end{bmatrix}$, find row 2 of AB.



A [1, -12]

B **[8, -5]** ✓

C [-5, 8]

D [4, -6]

WORKED SOLUTION & VERIFICATION

$(AB)_{21} = -1(-4) + (-2)(-2) = 8$, $(AB)_{22} = -1(-1) + (-2)(3) = -5$.